

TrES-1b

R-0226

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-1_250617 · created 2026-07-17T02:41:42+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460843.7943 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1307 +/- 0.0099
Transit depth (Rp/Rs)^2	1.71 +/- 0.26 [%]
Orbital Inclination (inc)	88.58 +/- 0.68
Airmass coefficient 1 (a1)	1.026 +/- 0.011
Airmass coefficient 2 (a2)	-0.025 +/- 0.016
Scatter in the residuals of the lightcurve fit is	1.07 %
Best Comparison Star	#2 - [520, 386]
Optimal Method	PSF photometry
Transit Duration (day)	0.1005 +/- 0.0029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.1307 ± 0.0099 vs 0.1358 (deviation 0.38σ)
Duration fit vs published	0.1005 d vs 0.1042 d (deviation 0.94σ)
Mid-time O–C	1.2 min
Depth SNR	9.8
Residual scatter	1.07 %

Light curve

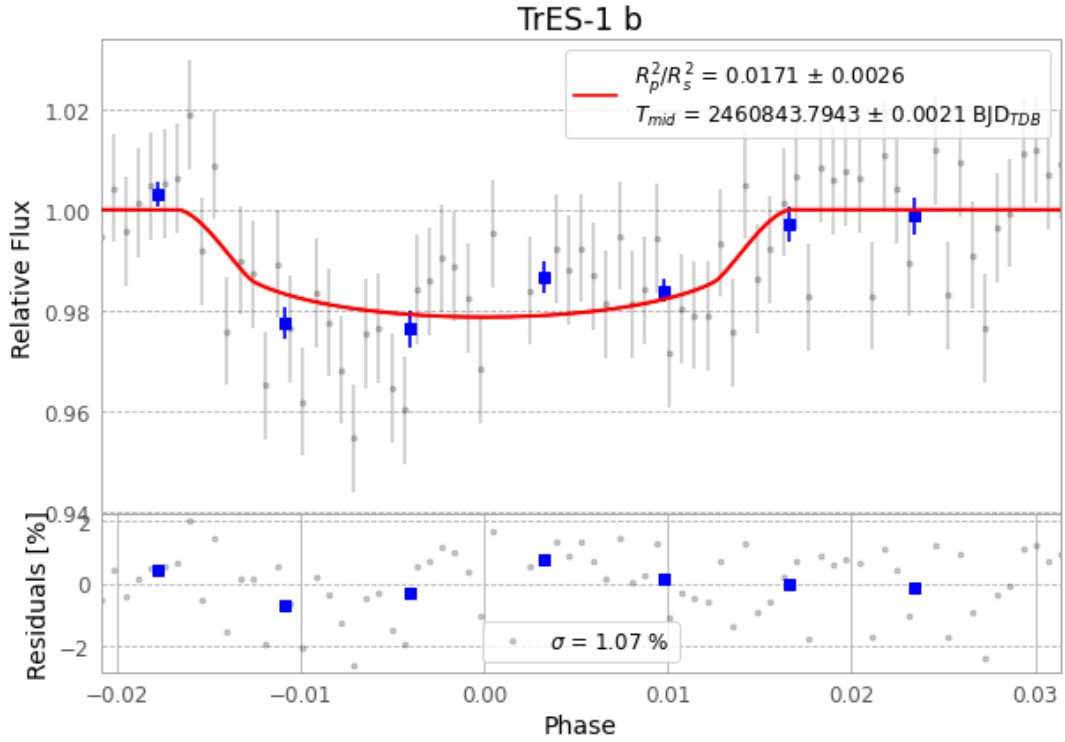


Figure 1 — FinalLightCurve_TrES-1 b_2025-06-16.png (SHA-256 1ee3c9a0dde1c7e4...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [346, 204], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 739a18d9f55f0317...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 825eb7e

Data provenance

74 FITS frames (49 MB), TRES-1250617052715.FITS → TRES-1250617091515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-250617.log (95bfea9dce5e...), FinalLightCurve_TrES-1 b_2025-06-16.png (1ee3c9a0dde1...), FinalLightCurve_TrES-1 b_2025-06-16.csv (61be6c6cfb54...)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 02:41Z.